DATA ANALYTICS COURSEWORK | PYTHON

ACH Transaction Matching

Normalize payment descriptions, suggest processor matches, and send uncertain records to a review queue.

1,000
transactions

10
processor labels

194
flagged for review

Objective

Turn inconsistent ACH descriptions into useful processor and customer summaries while avoiding forced assignments when the text match is uncertain.

Workflow

The Python workflow standardizes case and delimiters, removes transaction IDs and common banking tokens, expands known abbreviations, and compares the remaining tokens against processor aliases. A custom token-set similarity score uses Python's `SequenceMatcher`. A heuristic 80-point boundary separates confident suggestions from records routed to manual review.

Recorded Results

Status	Transactions	Dollar amount
Confident match (heuristic threshold)	806	$3,697,542.35
Flagged for manual review	194	$400,550.92

Toast has the highest recorded confident-match volume ($415,515.99), followed by Fiserv / First Data ($403,236.36). Customer aggregates rank accounts by volume, transaction count, primary processor, and number of distinct processors.

Operational Safeguard

> **The score is not a validated probability.** The cutoff is based on the score distribution in this dataset, not labeled ground truth. Precision, recall, and false-match rates are unknown. Low-scoring suggestions remain unconfirmed and should be reviewed by a person.

Reproduction

Install dependencies with `python -m pip install -r requirements.txt`, then run `python analysis.py`. The script reads the bundled CSV beside itself and prints its transaction review and aggregation tables.

Coursework workflow on a supplied dataset. Not a payment processor, reconciliation system, or autonomous financial control.